Novosibirsk State University
*THE REAL SCIENCE

TELECOMMUNICATION CUSTOMERS CHURN PREDICTION USING HYBRID NEURAL NETWORK AND INTEGRATED INTERPRETABLE ML(XAI) METHODS.

APPLIED MATHEMATICS AND INFORMATICS — ARTIFICIAL INTELLIGENCE AND BIG DATA ANALYSIS.

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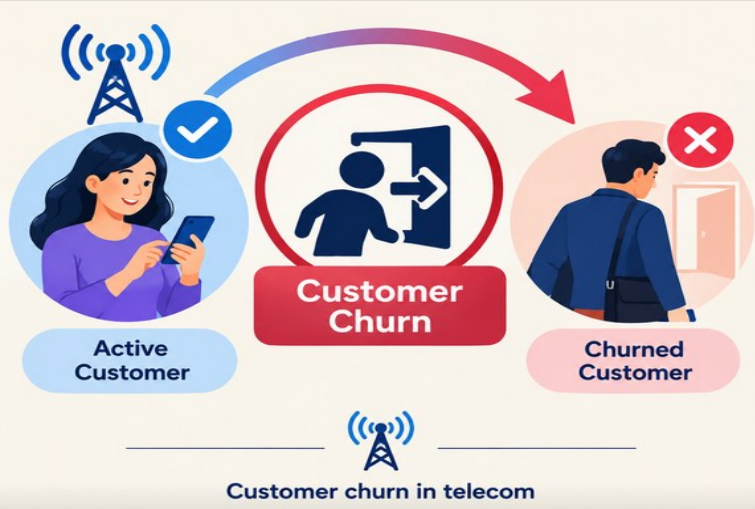
RELEVANCE

- **Relevance:**

- In a hypercompetitive telecom environment:

→ High risk of customer churn.

→ Acquisition costs > retention costs.





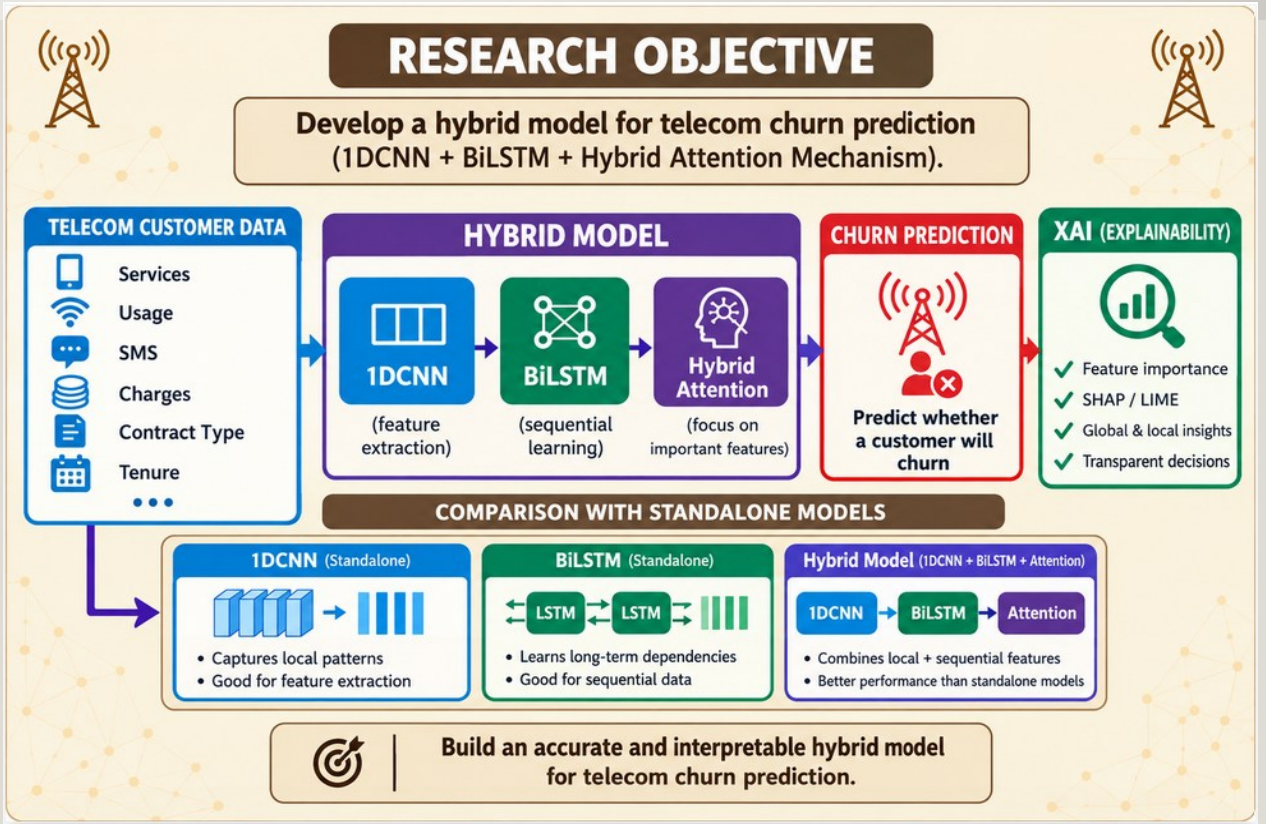
OBJECTIVE OF THE RESEARCH

- **Research objective:**

→ Develop a hybrid model:
(1DCNN + BiLSTM +
hybrid attention mechanism).

→ Comparison with standalone models:
(1D CNN, BiLSTM).

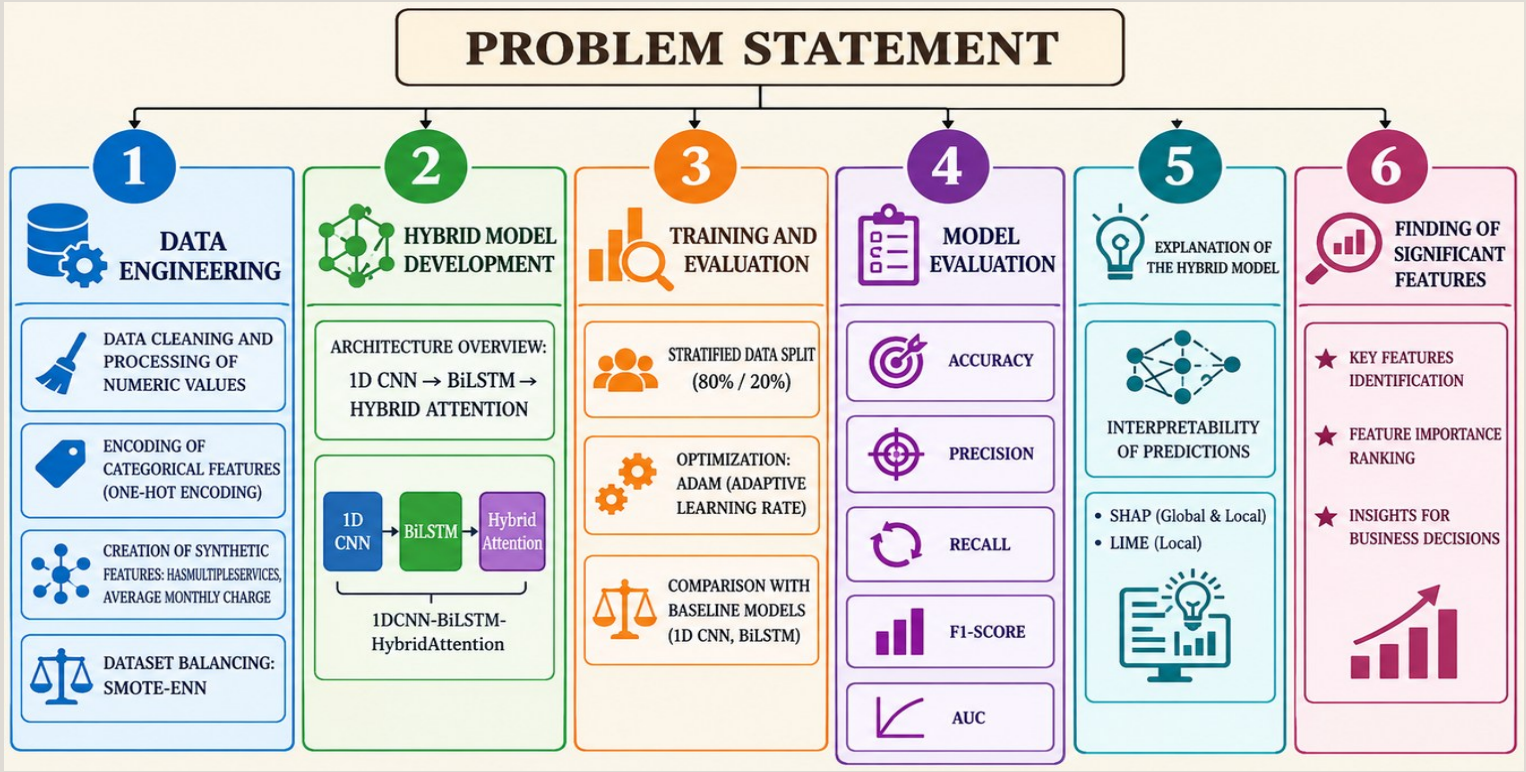
→ Explain the "*black-box*" hybrid model prediction.



PROBLEM STATEMENT

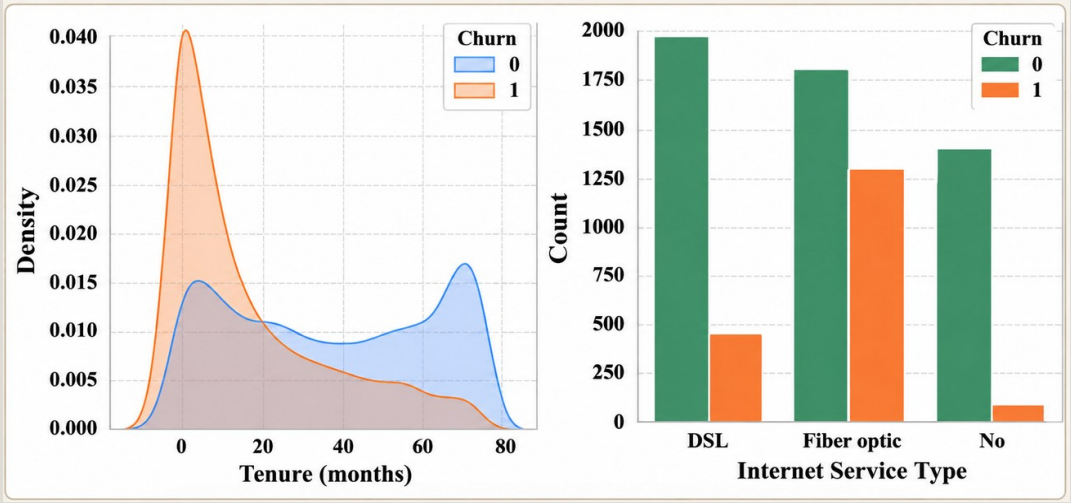
- **To achieve the stated goal, the following parts of tasks were implemented:**

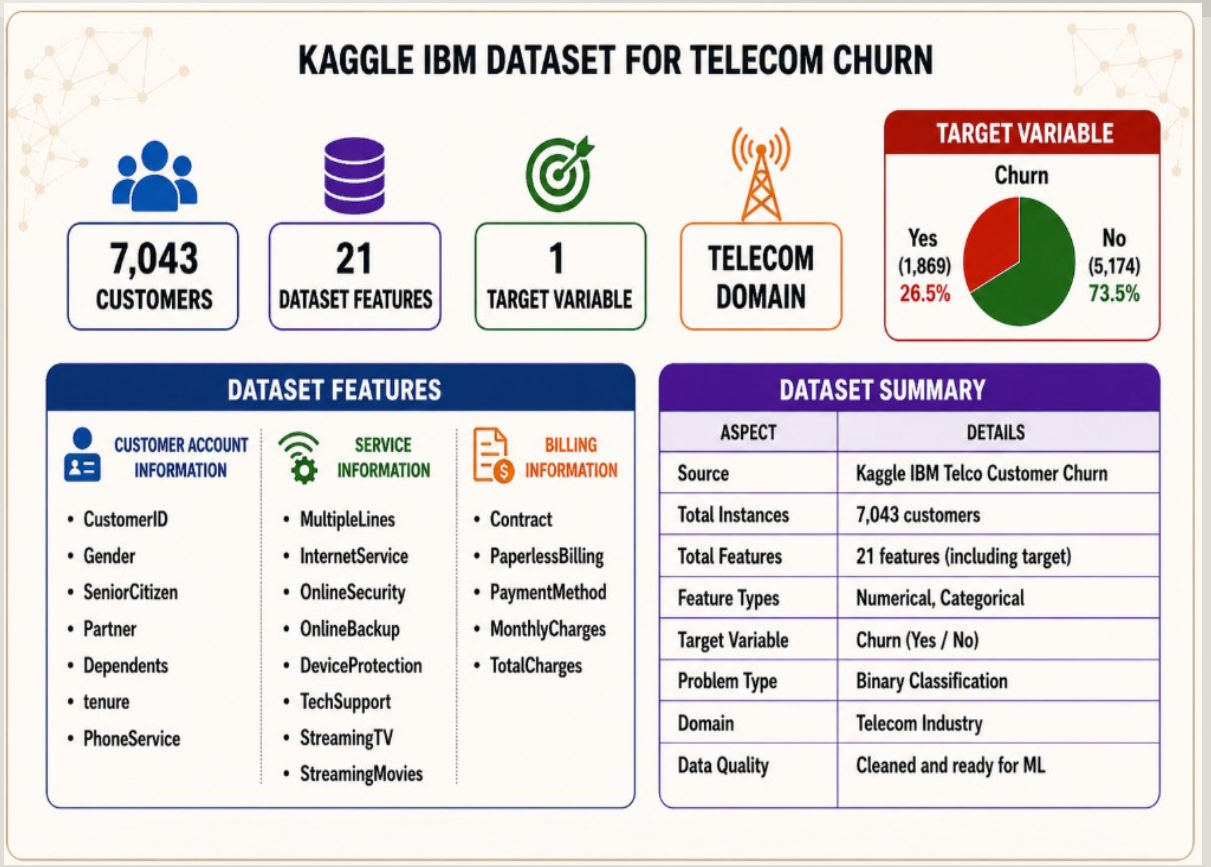
1. Data Engineering.
2. Hybrid model development.
3. Training of the hybrid model.
4. Hybrid model evaluation.
5. Hybrid model explanation.
6. Finding of significant features.



DATA PREPARATION AND FEATURE ENGINEERING

- Overview of the dataset:
  (IBM Telco, 7043 customers, 21 features).
- Customers: (*0–5 months*) — new(churn).
  > 65 months — loyal (not churn).
- *Fiber-optic internet* users are more prone to churn.





DATA PREPARATION AND FEATURE ENGINEERING (CONTINUED…)

- **Filling null-values, OHE(Categorical), Scaling(Numerical).**
- **Synthetic(engineered) features:**

→ HasMultipleServices:

$$HasMultipleServices = \sum_{j=1}^{ServiceCount} (Service_j = Active)$$

→ AverageMonthlyCharge:

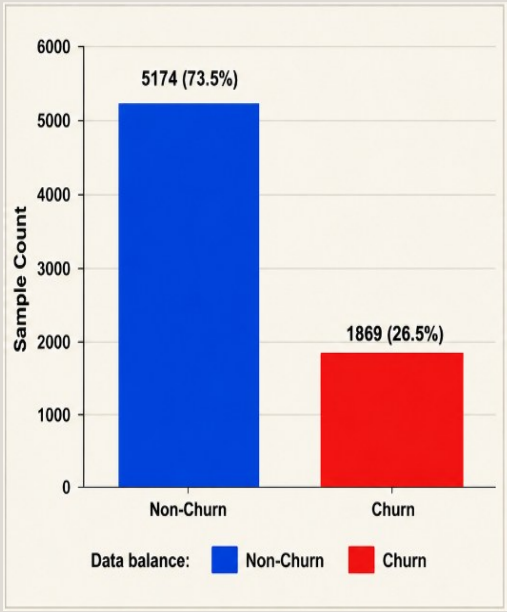
$$AverageMonthlyCharge = \frac{TotalCharges}{tenure + 1}$$

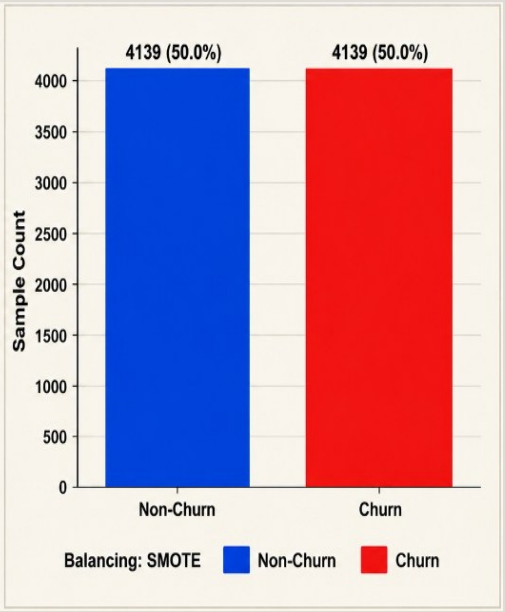
- **Dataset balancing**:

→ SMOTE

(73.5%→50.0% Non-Churn ,

26.5%→ 50.0% Churn).



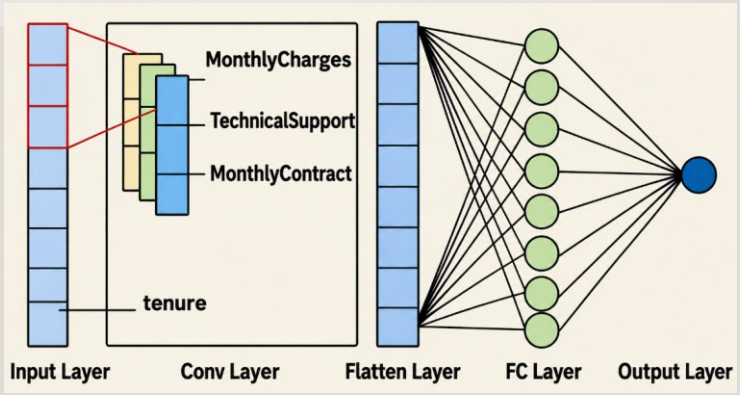


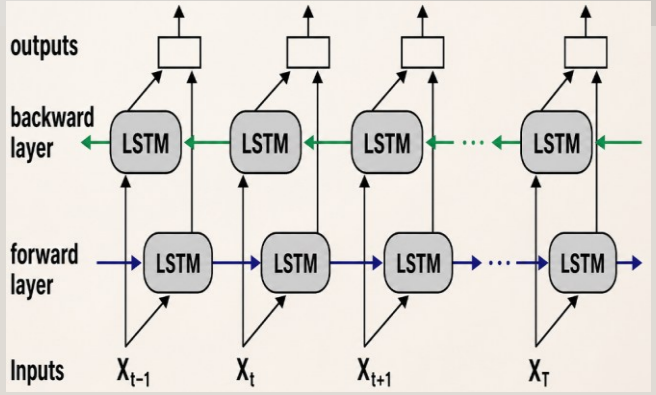
CustomerID	Contract	InternetService	PaymentMethod	Monthly Charges
A001	month-to-month	Fiber	Electronic check	$50.000
A002	one-year	DSL	Mailed check	$75.273
A003	two-years	No	Bank transfer (automatic)	$56.355

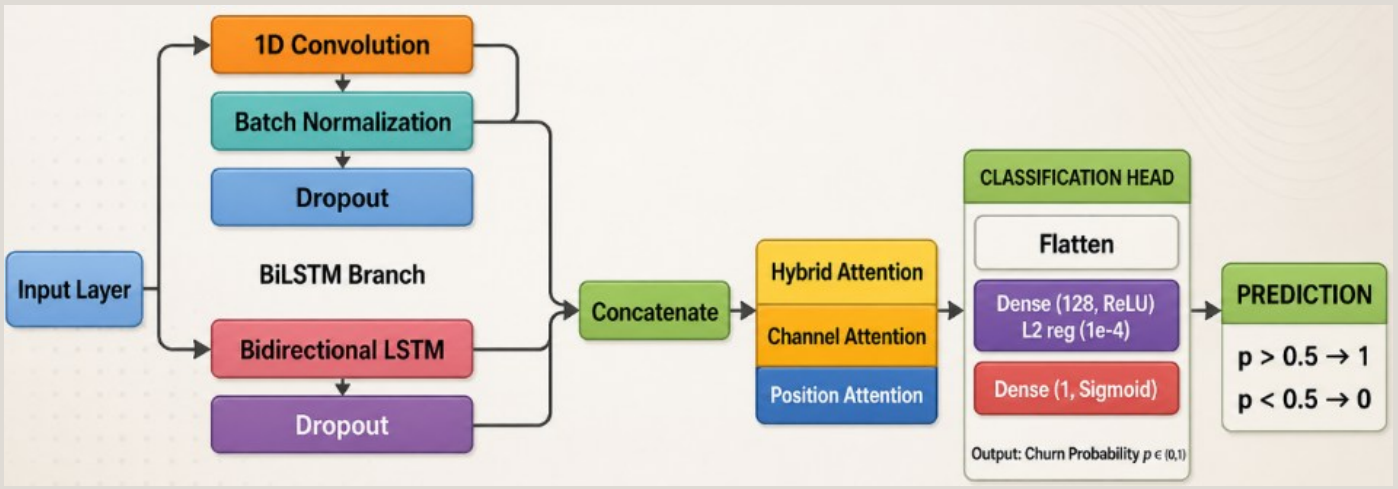
Encoding	Normalizing Numerical Features			Feature Matrix			
Categorical Features	Tenure	Monthly Charges	Total Charges	f1	f2	f3	...
One-Hot Encoding	19.43	29.35	3.18	29.45	64.45	63	...
	69.07	56.95	6.0	23.07	53.35	82	...
	49.221	60.23	0.0	49.82	53.35	78	...
	Total/Avg	63.228	29.25	53.55	53.55	55	...

IMPLEMENTED DEEP LEARNING ARCHITECTURES

- **1D CNN** → detection of local patterns along the feature axis.
- **BiLSTM** → Capture sequential patterns from both past and future contexts.
- **Proposed hybrid model:** 1D CNN + BiLSTM + hybrid attention (channel + positional).





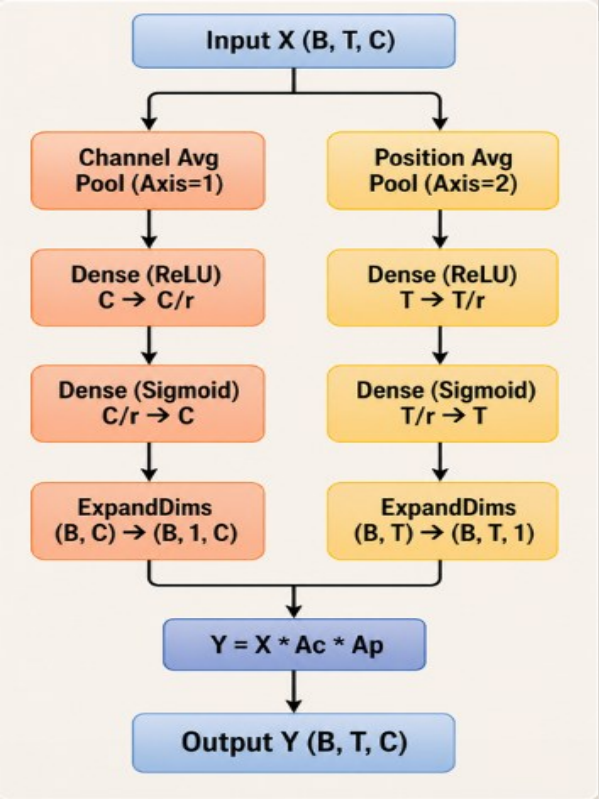


HYBRID ATTENTION MECHANISM (KEY NOVELTY)

- Input data: concatenated 1D CNN - BiLSTM features.
- Components → (channel attention + positional attention).
- Channel attention → which features are important?
- Positional attention → where are they important?
- Dynamically extract and weight significant features.
- Simplified mathematical formula: $Y = X * A_c * A_p$

  Where ***X***— combined feature map, ***Y*** — refined feature map,

  Ac — channel attention weights, ***Ap*** — position attention weights.
- Output data: recalibrated features.



HYPERPARAMETER CONFIGURATIONS OF THE EVALUATED MODELS.

Model	Category	Parameter	Specification / Value
CNN	Architecture	Layers & Activation	1× Conv1D (64 filters, kernel size 3, ReLU) + 1× BatchNormalization + 1× Conv1D (128 filters, kernel size 3, ReLU) + BatchNormalization
		Pooling	MaxPooling1D (pool size = 2, Dropout: 0.2) + GlobalAveragePooling1D (Dropout: 0.3)
	Training	Optimization & loss	Adam, Binary Crossentropy
BiLSTM	Architecture	Recurrent Layers	2× Stacked Bidirectional LSTM (1st layer: 64 units, 2nd layer: 32 units)
		Regularization	L2 = 0.001, Dropout: 0.3 (applied after each BiLSTM layer)
	Training	Optimization	Adam, Binary Crossentropy
CNN-BiLSTM-HybridAttention	Architecture	CNN Branch	64 filters, kernel size 3 (padding="same"), ReLU, L2 = 1e-4, Dropout: 0.3, Batch Normalization
		LSTM Branch	Bidirectional LSTM (64 units, return_sequences=True), L2 = 1e-4, Dropout: 0.3
		Attention Module	HybridAttention1D (Reduction ratio = 8, Dropout: 0.4, L2 = 1e-4)
		Classification Head	Flatten, Dense layer (128 units, ReLU, L2 = 1e-4, Dropout: 0.4), Output: Dense(1, Sigmoid)
	Training	Optimization	Adam (lr = 0.0003, weight decay = 1e-6), Focal Loss (γ=2.5, α=0.75)

RESULTS OF THE COMPUTATIONAL EXPERIMENT.

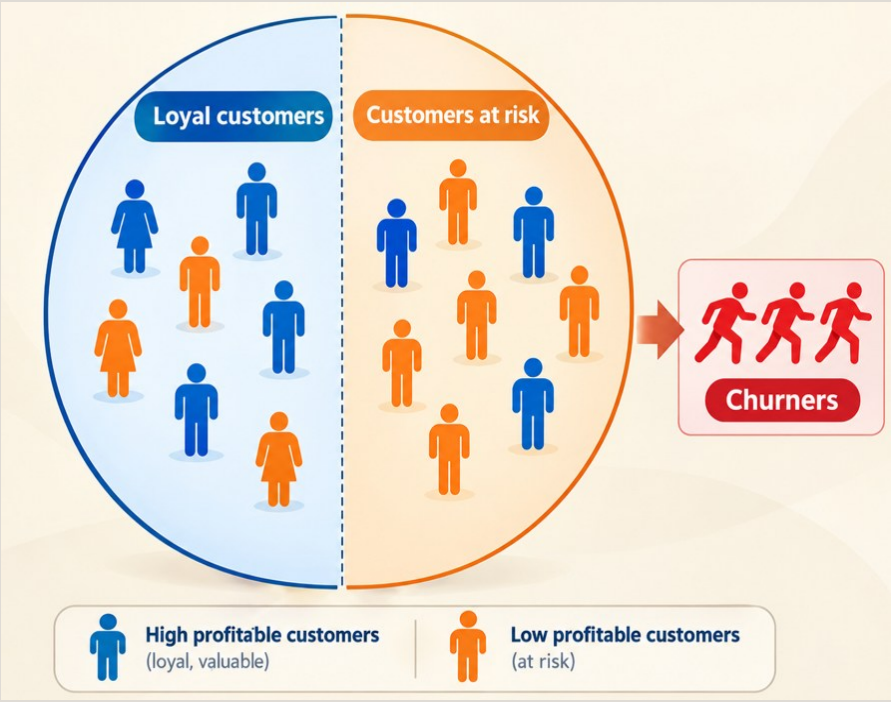
- **Performance metrics comparison:**

→ (Accuracy, Precision, Recall, F1-Score, AUC).

- **The hybrid model shows best results:**

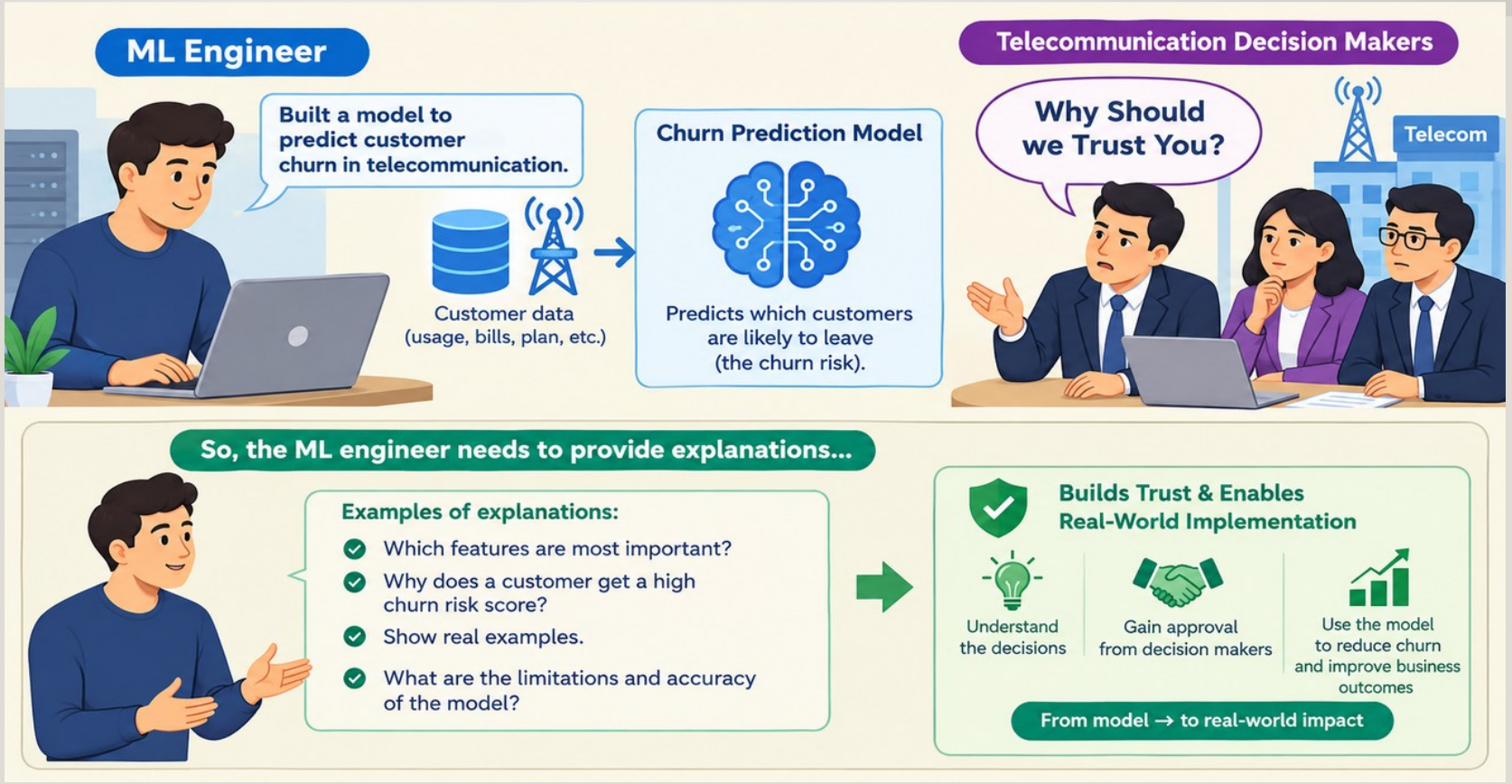
→ (Accuracy 82.04%, F1-Score 67.52% and AUC 0.86).

Model	Accuracy	Precision	Recall	F1-Score	AUC
CNN	77.36%	54.96%	80.16%	65.21%	0.86
BiLSTM	62.60%	41.07%	94.91%	57.33%	0.85
CNN-BiLSTM-HybridAttention	82.04%	64.78%	70.51%	67.52%	0.86



WHY WE SHOULD WE TRUST YOU?

Explain for us?



PLAN OF EXPLAINABILITY AFTER TRAINING (WHY?)

Integrate four explainable methods:

→ **SHAP**, Partial Dependence Plot (**PDP**),

Differential Accumulated Local Effect (**DALE**) and **LIME**.

- Global Explainability:

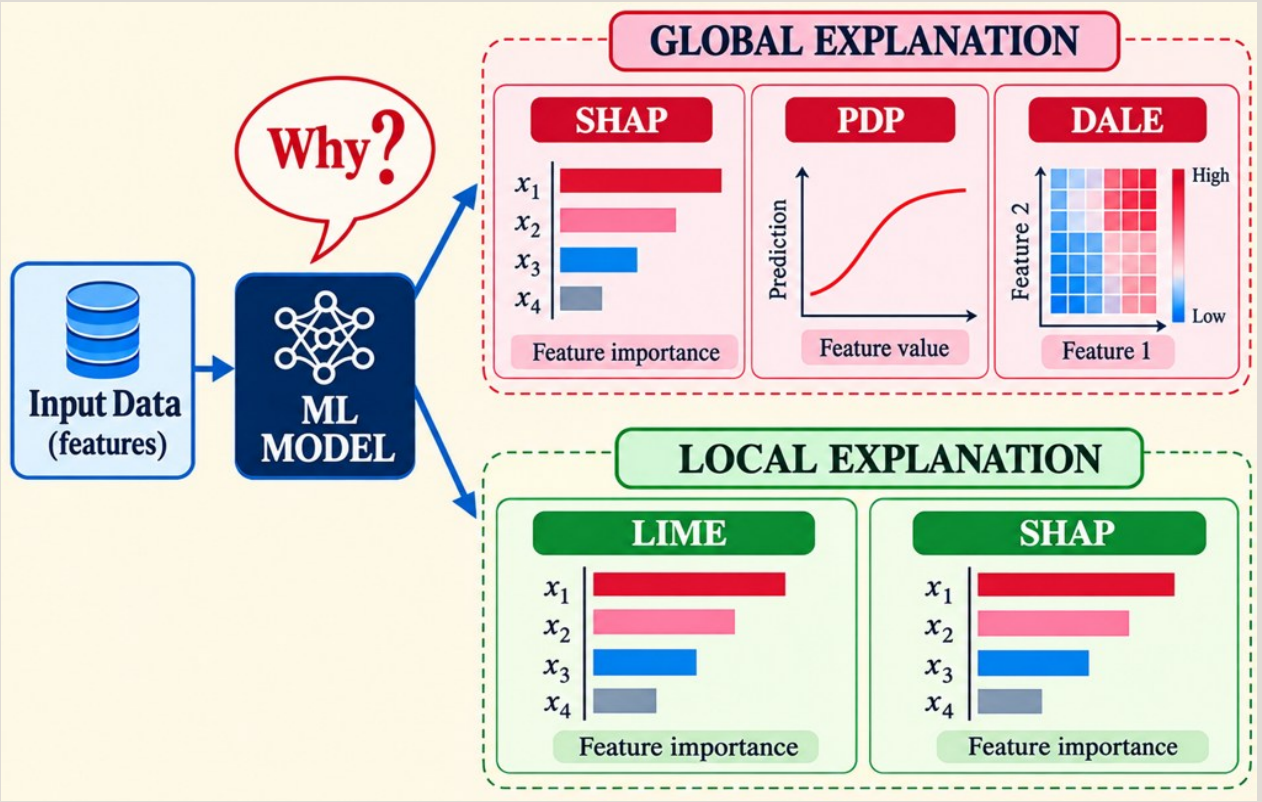
→ **SHAP**(Summary plot, and Interaction plot),

PDP and **DALE**.

- Local Explainability:

→ **SHAP**(waterfall plot, force plot, decision plot),

LIME(Bar chart, Decision plot).

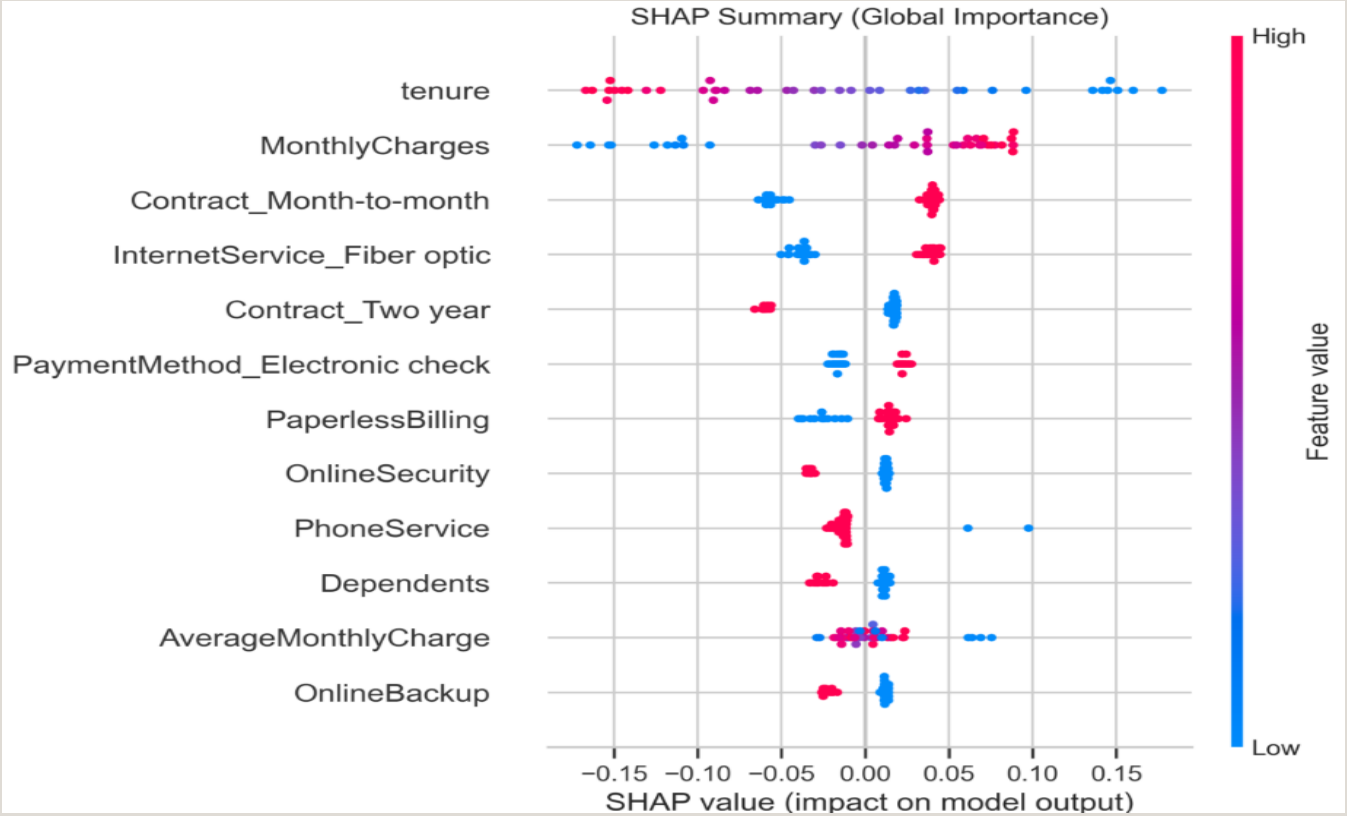


GLOBAL EXPLAINABILITY USING SHAP (SHAP SUMMARY PLOT)

SHAP Summary plot:

→ Explains the entire model's behavior across all customers.

- Red flagged colors:
  → Churn (Exit). High.
- Blue flagged colors:
  → Not-churn (Stay). Low.
- Top four important features:
  → Tenure, MonthlyCharges, Month-to-Month contract, Fiber-Optic Internet service.

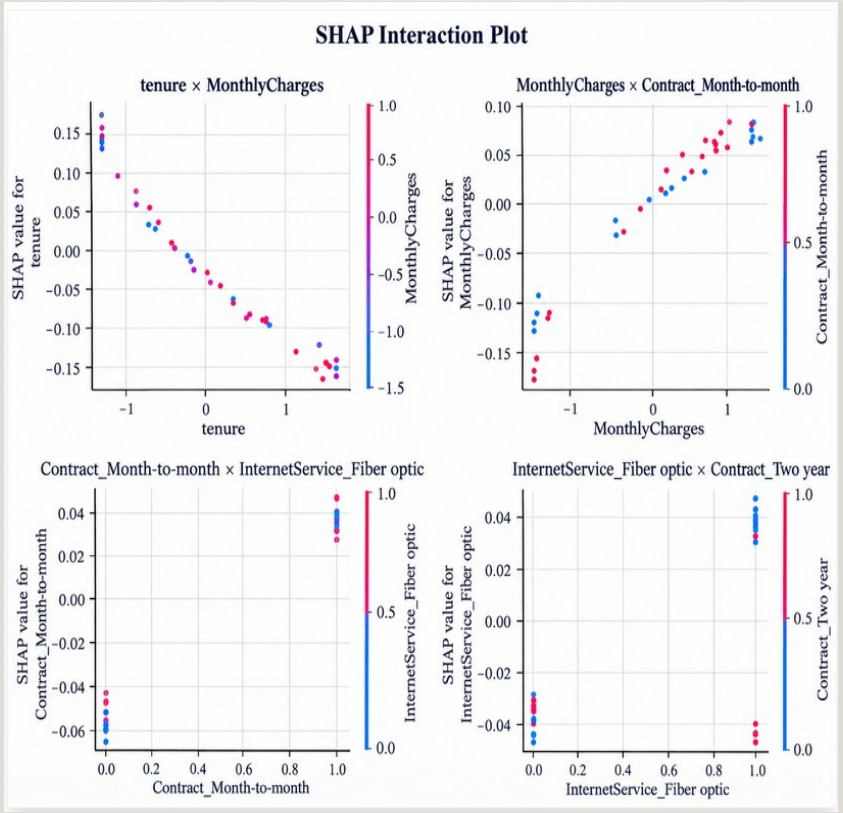


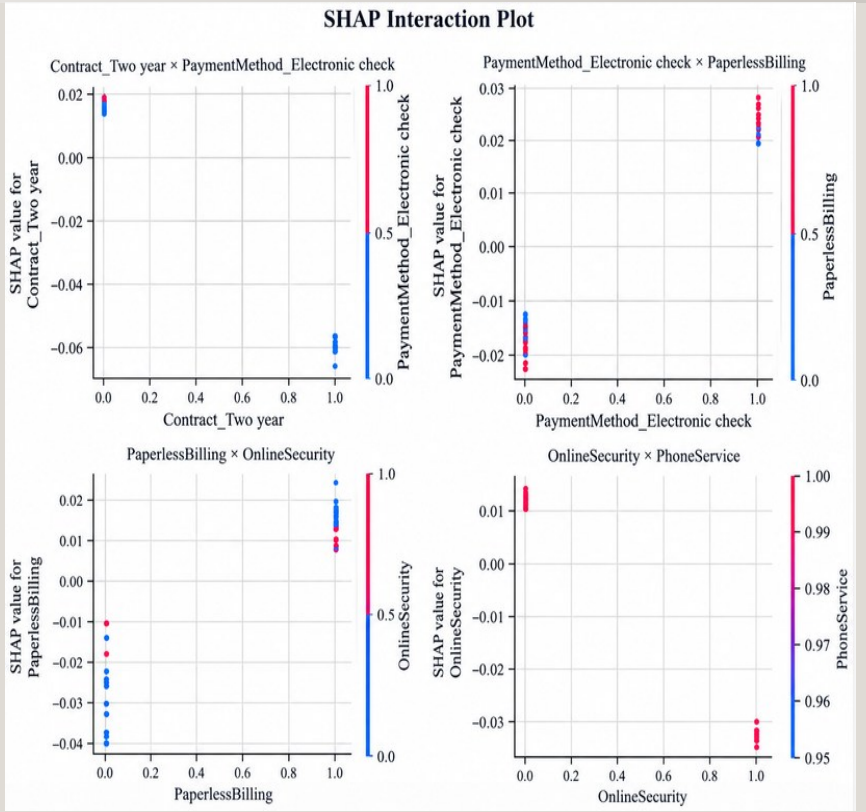
GLOBAL EXPLAINABILITY USING SHAP (SHAP INTERACTION PLOT)

SHAP Interaction Plot:

- The effect of a single feature on the prediction, conditioned on the value of a second feature.

- Shorter customer *tenure* strongly increases churn risk, and this risk is slightly amplified for those with higher monthly charges.

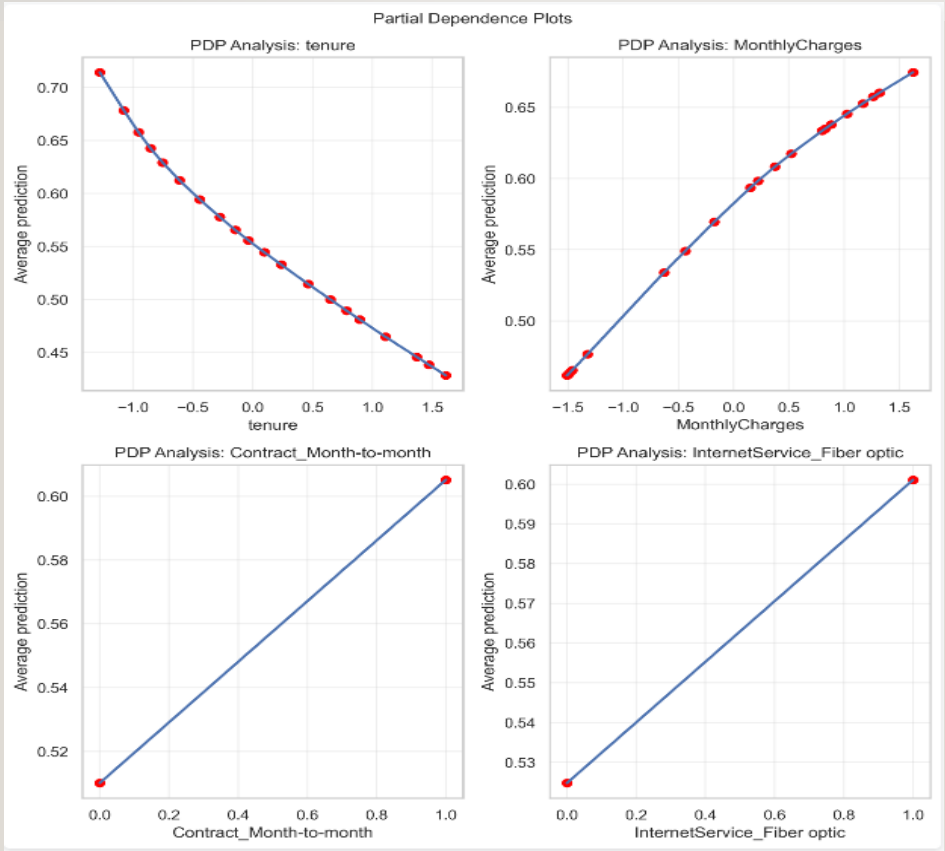


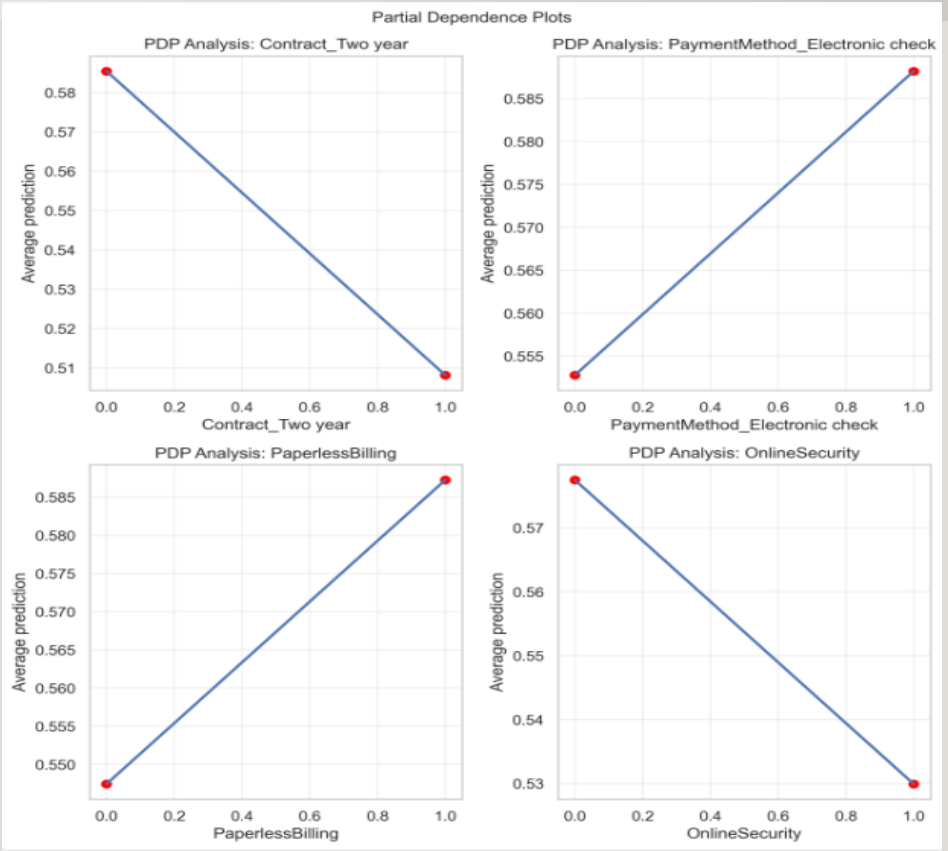


GLOBAL EXPLAINABILITY USING PDP(PARTIAL DEPENDENCE PLOT)

Partial Dependence Plot:

- Average prediction of a single feature over the entire customer base.





CONCLUSION AND OUR NEXT WORK

- **Conclusion:**

→ The hybrid model outperforms standalone 1D CNN and BiLSTM models.

→ Global key features: *tenure, monthly charges, month-to-month contract, and fiber optic internet service*.

- **Next work:**

→ *Global Explainability*:

→ Differential Accumulated Local Effect (DALE)

→ *Local Explainability*:

→ **SHAP**(waterfall plot, force plot, decision plot), **LIME**(Bar chart, Decision plot).

THE END!

Thank you !!!

